# PROMPT ADHERENCE IN TEXT-TO-IMAGE DIFFUSION MODELS: PROMPT STYLING, REFERENCE-GUIDED EVALUATION, AND CROSS-ATTENTION ANALYSIS

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## ABSTRACT

Text-to-image diffusion models can generate visually convincing images while failing to preserve the semantic constraints of a natural-language prompt. This work proposes an experimental framework for investigating prompt adherence through prompt styling, reference-image comparison, quantitative evaluation, and cross-attention analysis. The framework separates observable image correctness from the internal attention patterns associated with prompt tokens. A focused parameter experiment uses RealVisXL V5.0 with the prompt "A red ball; A blue cube," three classifier-free guidance (CFG) scales, three denoising-step settings, and three random seeds, producing 27 images. Object detection, image-text similarity, reference-image similarity, and human judgments are treated as complementary evaluation signals rather than interchangeable measures. A normalized combined score is proposed, while token-level cross-attention maps are used for exploratory mechanistic analysis. The study is designed as a constrained pilot: results, score weights, and conclusions must be determined from the recorded experiments rather than assumed in advance.

***Index Terms*—** Text-to-image generation, prompt adherence, prompt styling, diffusion models, reference-image similarity, classifier-free guidance, cross-attention

## 1. INTRODUCTION

Text-to-image diffusion models have advanced considerably in generating images from natural-language descriptions. Nevertheless, a visually appealing output does not necessarily satisfy every condition in the prompt. Models may omit requested entities, associate attributes with the wrong objects, or produce incorrect spatial relationships.

These failures are especially important in compositional prompts, where multiple entities and attributes must be represented simultaneously. For example, the prompt "A red ball; A blue cube" requires the presence of two distinct objects and the correct association between each object and its color. An image containing a red object and a blue object is not sufficient if the shapes and colors are incorrectly paired.

Prompt adherence can be investigated at several complementary levels. First, prompt styling can alter the organization and explicitness of the textual conditioning without changing its intended meaning. Second, reference-image comparison can measure visual similarity to a chosen target. Third, object detectors and human assessments can evaluate whether requested entities and attributes are actually present. Finally, cross-attention maps can provide a view of how image locations attend to prompt tokens during generation.

These measurements answer different questions. A reference image may be visually similar to a generated image even when a small semantic constraint is violated. An object detector may identify the requested objects without verifying their color assignments. Cross-attention intensity may reveal where a token receives attention, but it does not independently establish that the generated image is correct.

This paper develops a unified framework that distinguishes these evaluation signals and investigates their relationship with generation parameters. The experimental plan has four connected components: prompt styling, reference-guided evaluation, parameter sensitivity analysis, and cross-attention-based mechanistic investigation.

The main contributions are:

- A controlled prompt-styling framework that aims to preserve semantic constraints while varying their textual presentation.
- A reference-guided evaluation procedure that complements object-level and human adherence judgments.
- A proposed combined score for summarizing complementary measures of image adherence.
- A controlled CFG and denoising-step experiment using 27 generations of a fixed prompt.
- A cross-attention analysis procedure for visualizing token-specific spatial attention and examining its relationship with observed adherence.

The present parameter experiment is a pilot using one prompt. Its results should not be interpreted as evidence of generalization to all prompt types.

## 2. RELATED WORK

### 2.1. Diffusion Models and Text Conditioning

Denoising diffusion probabilistic models generate samples by learning to reverse a gradual noising process [1]. Latent diffusion models move the process into a compressed latent space, reducing computational requirements while supporting high-resolution image synthesis [3].

In text-conditioned generation, a text encoder produces representations of the prompt that interact with the image-generation network. The resulting image depends on the learned model, textual conditioning, random initialization, scheduler, and inference settings. Consequently, prompt adherence cannot be understood solely from the final text string or the visual quality of the output.

The experiments in this paper use the SDXL-based checkpoint RealVisXL V5.0. The checkpoint is kept fixed so that the study focuses on inference-time conditions rather than comparisons between model architectures.

### 2.2. Prompt Formulation and Attribute Binding

Prompt formulation is a practical means of communicating semantic requirements to text-to-image models. Different word orders, explicit attribute associations, and line-separated descriptions may influence the generated composition. However, a textual reformulation is not guaranteed to improve adherence, and stylistic changes can inadvertently add, remove, or alter constraints.

Attribute binding concerns whether an attribute is associated with its intended entity. For example, "red ball" and "blue cube" specify two separate color-object associations. Evaluating this property requires more than determining whether the image contains red and blue regions or whether both object classes are detectable.

The present work treats prompt styling as an experimental variable and requires semantic equivalence checks before comparing formulations.

### 2.3. Classifier-Free Guidance

Classifier-free guidance combines conditional and unconditional model predictions to control the influence of conditioning information during sampling [2]. CFG is commonly adjusted to trade off conditioning strength against other aspects of image generation. Its relationship with prompt adherence need not be monotonic: a larger CFG value does not guarantee more accurate attributes or better composition.

The experiment compares CFG scales of 3, 7.5, and 12 under multiple denoising-step settings.

### 2.4. Reference-Image Similarity

Image-text and image-image similarity models provide useful signals for comparing visual content. CLIP learns aligned image and text representations that can be used for similarity-based evaluation [4]. A reference image can therefore provide a target for comparing generated appearance and composition.

However, a high reference-image similarity score does not guarantee exact object counts, correct color-object associations, or full prompt adherence. The reference-image signal must be interpreted alongside object-level and human assessments.

In this study, a reference image is an evaluation target. Unless a separate reference-conditioned generation method is explicitly implemented, the image is not treated as a training example that updates the diffusion model's weights.

### 2.5. Cross-Attention and Mechanistic Analysis

Cross-attention allows spatial image features to attend to representations of textual tokens. Visualizing cross-attention can reveal which image regions are associated with particular tokens during the denoising trajectory. Related work has explored attention-based localization and interventions in text-to-image diffusion models [6].

Attention maps are useful for exploring possible mechanisms behind prompt adherence, but they are not ground-truth explanations. High attention on a token does not necessarily mean that the corresponding attribute is rendered correctly. The present work therefore treats cross-attention maps as a mechanistic analysis tool, not as a replacement for image-level evaluation.

## 3. RESEARCH OBJECTIVES AND QUESTIONS

The framework addresses four questions.

**RQ1: Prompt styling.** Does changing the textual organization of a semantically equivalent prompt change the observed adherence of generated images?

**RQ2: Reference-guided evaluation.** How does similarity to a chosen reference image relate to object-level and human-rated adherence?

**RQ3: Parameter sensitivity.** How do CFG scale and denoising-step count relate to detection scores and observed attribute binding for a fixed prompt?

**RQ4: Mechanistic analysis.** How do token-level cross-attention maps vary across generation settings, and do these patterns correspond to differences in observed image adherence?

The parameter experiment directly investigates RQ3. The remaining questions define the broader framework and should only be described as empirically answered if their corresponding experiments have actually been completed.

## 4. PROPOSED EXPERIMENTAL FRAMEWORK

### 4.1. Overview

The overall workflow is illustrated in Fig. 1. It separates prompt formulation, image generation, reference comparison, adherence scoring, and mechanistic analysis.

### 4.2. Prompt Styling

Prompt styling is investigated by reformulating a fixed set of semantic constraints without intentionally adding new requirements. Candidate formulations may include:

- **Baseline:** the original natural-language prompt.
- **Entity-first:** the requested entities are stated clearly and directly.
- **Attribute-explicit:** each attribute is placed immediately beside its intended entity.
- **Line-separated:** each entity-attribute pair is placed on a separate line.
- **Concise:** redundant wording is removed without deleting any constraint.

For the parameter experiment, the selected prompt is:

A red ball
A blue cube

This wording is fixed across all 27 parameter configurations. It is not modified separately for individual CFG values or denoising-step counts.

The styling comparison and the parameter sweep answer different questions. When comparing styles, model settings and random seeds should be held fixed. Once a style has been selected using a documented criterion, it should remain unchanged during the parameter experiment.

If time constraints prevent a complete comparison of multiple styles, the paper must describe prompt styling as a proposed or partially completed stage rather than claim that an optimal style was experimentally established.

### 4.3. Reference Image Selection

A reference image is selected to represent the intended visual composition. For the attribute-binding task, it should clearly depict a red ball and a blue cube. The reference image and its source should be documented, and the same reference should be used across the compared generation settings.

The reference image is used for evaluation rather than assumed to be a training image. The main diffusion checkpoint remains fixed. If a future experiment uses reference conditioning, fine-tuning, or another image-guided generation mechanism, that intervention must be described as a separate experimental condition.

Reference-image similarity is complementary to semantic adherence. A reference can guide the desired appearance, but a pixel-level or embedding-level match is not a definitive correctness test.

### 4.4. Image Generation Model

The model used in the parameter experiment is `SG161222/RealVisXL_`
The prompt, model checkpoint, image resolution, scheduler, and other inference settings are held constant unless explicitly identified as experimental variables.

The chosen prompt is:

A red ball
A blue cube

The intended semantic constraints are:

1. A ball must be present.
2. The ball must be red.
3. A cube must be present.
4. The cube must be blue.

### 4.5. CFG and Denoising-Step Experiment

The parameter experiment uses a full-factorial design. The CFG scale has three levels, the number of denoising steps has three levels, and three random seeds are used for each configuration.

**Table 1**. Fixed experimental settings.

| Parameter | Values |
|---|---|
| Model | RealVisXL V5.0 |
| CFG scale | 3, 7.5, 12 |
| Denoising steps | 20, 30, 40 |
| Random seeds | 42, 123, 2026 |
| Resolution | $1024 \times 1024$ |
| Prompt | A red ball; A blue cube |
| Unique CFG-step pairs | 9 |
| Images per pair | 3 |
| Total images | 27 |

The total number of generations is

$$N = N_{\mathrm{CFG}} N_{\mathrm{steps}} N_{\mathrm{seeds}} = 3 \times 3 \times 3 = 27. \qquad (1)$$

The same seeds are reused across parameter configurations. This matched-seed design supports comparisons under

**Reserved space for the complete research pipeline diagram**

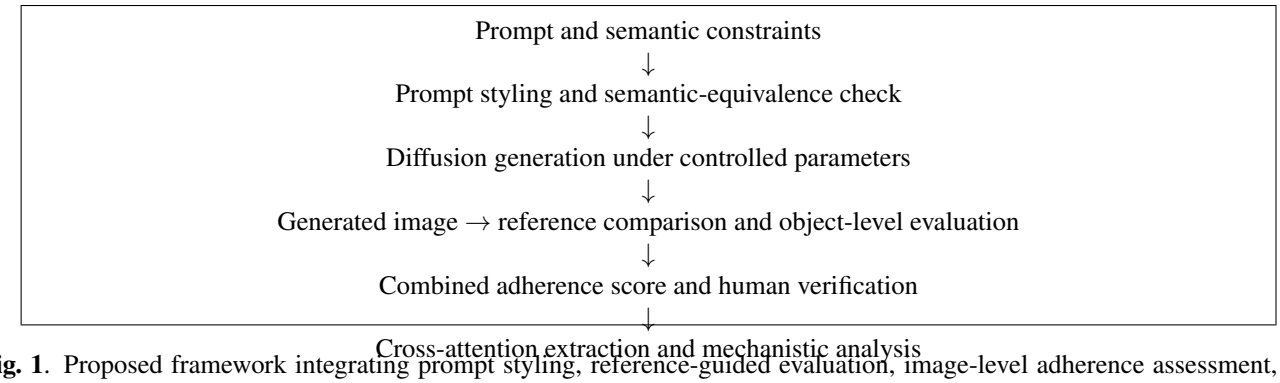



**Fig. 1**. Proposed framework integrating prompt styling, reference-guided evaluation, image-level adherence assessment, and cross-attention analysis.

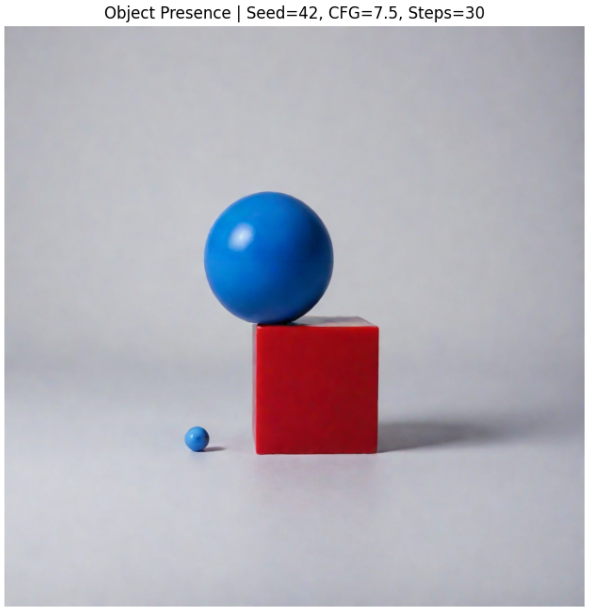



**Fig. 2**. Reference image defining the target composition for reference-guided evaluation.

repeated initial random conditions. However, changing inference settings can change the entire sampling trajectory, so matching seeds does not make the generated images identical apart from one isolated parameter.

### 4.6. Controlled Variables

For the parameter experiment, the following variables are controlled:

- Prompt text and line formatting.
- Model checkpoint.
- Reference image used for evaluation.
- Output resolution.
- Scheduler and other inference settings.
- Negative prompt, if used.
- Software environment and evaluation procedure.

Only CFG scale, denoising-step count, and random seed vary in the 27-image experiment. All outputs, including failures, should be retained with metadata.

## 5. EVALUATION METHODOLOGY

### 5.1. Evaluation Dimensions

Prompt adherence is not identical to visual quality. The evaluation therefore separates the following dimensions:

1. **Object presence:** whether both requested objects are present.
2. **Attribute binding:** whether the ball is red and the cube is blue.
3. **Reference similarity:** whether the generated image is visually similar to the chosen reference.
4. **Overall adherence:** whether the generated image satisfies the prompt's semantic constraints as a whole.
5. **Mechanistic attention:** how the spatial attention associated with relevant text tokens changes during denoising.

The first four dimensions concern image-level evaluation. The fifth concerns internal model behavior and should be analyzed separately.

### 5.2. Object Detection

OWL-ViT is used as a preliminary open-vocabulary detection tool [5]. The image is evaluated using the text queries "a red ball" and "a blue cube". For each query, the maximum returned detection score is recorded.

Let the scores be $S_b$ for the ball query and $S_c$ for the cube query. The weaker detection score is

$$S_{\min} = \min(S_b, S_c). \tag{2}$$

A provisional threshold $\tau = 0.1$ may be used to calculate a binary detection outcome:

$$D = \begin{cases} 1, & S_b \geq \tau \text{ and } S_c \geq \tau, \\ 0, & \text{otherwise.} \end{cases} \tag{3}$$

This threshold is an operational choice, not a validated measure of correctness. OWL-ViT scores are not guaranteed to be calibrated probabilities, and these queries do not independently prove that each color is associated with the correct shape. Consequently, detection scores are retained as one component of the evaluation.

### 5.3. Reference-Image Similarity

Let $I_g$ denote a generated image and $I_r$ the selected reference image. A pretrained image encoder maps each image to a feature vector:

$$\mathbf{z}_g = f(I_g), \qquad \mathbf{z}_r = f(I_r). \tag{4}$$

The cosine similarity is

$$S_{\text{ref}} = \frac{\mathbf{z}_g^\top \mathbf{z}_r}{\|\mathbf{z}_g\|_2 \|\mathbf{z}_r\|_2}. \tag{5}$$

A CLIP image encoder may be used for this purpose. The model identifier, preprocessing, and reference image must be held fixed across comparisons.

The similarity is normalized to the range $[0, 1]$ before it is combined with other components. One simple transformation is

$$R_{\text{ref}} = \text{clip}\left(\frac{S_{\text{ref}} + 1}{2}, 0, 1\right). \tag{6}$$

This transformation maps the theoretical cosine-similarity range to $[0, 1]$; it does not calibrate the score to a probability of adherence. If the chosen encoder's similarity is not cosine similarity, its normalization must be adapted and documented.

### 5.4. Human Evaluation

Automated metrics are supplemented by direct visual inspection. For every generated image, the reviewer records the following binary checks:

- Is the ball present?
- Is the cube present?
- Is the ball red?
- Is the cube blue?
- Are both objects and their attributes clearly distinguishable?

An overall adherence score can be assigned using a five-level rubric:

**Table 2**. Human-rated overall adherence rubric.

| Score | Description |
|---|---|
| 0 | Mostly incorrect or missing required content |
| 1 | Several major constraints are violated |
| 2 | Partial adherence; some constraints are satisfied |
| 3 | Nearly all constraints are satisfied |
| 4 | All required constraints are clearly satisfied |

The human score is normalized as

$$H = \frac{h}{4}, \qquad h \in \{0, 1, 2, 3, 4\}. \tag{7}$$

The rating rubric, number of reviewers, and any disagreements should be reported. If only one reviewer is used, the paper should not claim inter-rater reliability.

### 5.5. Combined Evaluation Score

The evaluation signals are complementary. A combined score is proposed to summarize the results, but its components must first be normalized to a common range.

Let:

- $D$ be the binary outcome that both requested objects exceed the provisional detection threshold;
- $A$ be the attribute-binding score, equal to 1 only when both color-object associations are visually correct and 0 otherwise;
- $R_{\text{ref}}$ be the normalized reference-image similarity;
- $H$ be the normalized human overall-adherence rating.

The proposed combined score is

$$\boxed{C = w_D D + w_A A + w_R R_{\text{ref}} + w_H H} \tag{8}$$

subject to

$$w_D + w_A + w_R + w_H = 1, \qquad w_i \geq 0. \tag{9}$$

The weights must be specified before interpreting the results. If no defensible calibration or validation data are available, equal weights may be used as a transparent exploratory baseline:

$$w_D = w_A = w_R = w_H = 0.25. \tag{10}$$

This equal-weight choice is not claimed to be optimal. Results should also be reported for the individual components

so that the combined score does not hide a failure in attribute binding behind a high reference-similarity value.

If manual attribute-binding labels have not yet been collected, $A$ must remain unfilled and the complete combined score must not be reported as though it had been measured. The score is a proposed evaluation framework until all required inputs are available.

Cross-attention statistics are not included directly in this combined score. They are used for the separate mechanistic investigation described in Section 6.

## 6. CROSS-ATTENTION MECHANISTIC ANALYSIS

### 6.1. Motivation

Image-level metrics can identify whether a generation appears to satisfy the prompt, but they do not directly show how textual tokens interact with spatial image features during generation. Cross-attention analysis provides a complementary view by visualizing the spatial distribution of attention associated with individual prompt tokens.

The analysis focuses on the tokens corresponding to "red", "ball", "blue", and "cube". The objective is to investigate whether the spatial attention patterns for each attribute and entity appear consistent with their intended associations.

### 6.2. Attention Map Extraction

In a cross-attention layer, image features provide queries, while text representations provide keys and values. A simplified attention expression is

$$\mathrm{Attn}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V, \tag{11}$$

where $Q$, $K$, and $V$ denote the query, key, and value representations, and $d$ is the key dimension.

The attention probabilities before multiplication by $V$ can be stored during the forward pass. For a particular text token, its attention weights can be reshaped into a spatial map corresponding to the image-feature resolution.

The attention-capture implementation should record the relevant layer, resolution, token index, denoising step, CFG setting, and random seed. Maps from different layers or resolutions should not be combined without documenting the aggregation procedure.

### 6.3. Token-Level Visualization

For each generation, attention maps should be extracted for the four relevant tokens. The maps are resized to the output-image dimensions for visualization and optionally overlaid on the generated image.

The visual comparison should examine whether the maps are concentrated around plausible regions and whether the attribute and object tokens attend to compatible regions. Such observations are qualitative and should not be interpreted as proof of causal binding.

### 6.4. Comparisons Across CFG Settings

For a fixed prompt, seed, and denoising-step count, attention maps are compared across CFG values. This makes it possible to inspect whether changing CFG is associated with differences in the spatial distribution of attention to the relevant tokens.

The maps should be generated using the same attention-extraction method and aggregation rules. If attention capture changes the execution path or numerical behavior, that possibility should be documented.

### 6.5. Comparisons Across Denoising Steps

For comparisons involving different total numbers of denoising steps, the analysis should align maps by relative denoising phase rather than comparing identical raw step indices. For example, early, middle, and late phases can be selected as fractions of each run's trajectory.

This avoids treating step 10 of a 20-step run as equivalent to step 10 of a 40-step run without justification. The same layer and spatial resolution should be used wherever possible.

### 6.6. Attention Statistics

As an exploratory measure, a token attention map $M_t$ may be normalized so that its nonnegative values sum to one:

$$\widetilde{M}_t(i,j) = \frac{M_t(i,j)}{\sum_{i,j} M_t(i,j) + \epsilon}, \tag{12}$$

where $\epsilon$ prevents division by zero.

Possible descriptive quantities include the spatial entropy

$$\mathcal{H}(M_t) = -\sum_{i,j} \widetilde{M}_t(i,j) \log\left(\widetilde{M}_t(i,j) + \epsilon\right) \tag{13}$$

and the change in normalized attention maps between configurations. These measures describe the concentration or change of attention; they do not directly measure semantic correctness.

If a spatial overlap score is used, the region definition must be stated in advance. For example, a manually annotated object region could be compared with the corresponding token map. Such an analysis requires reliable region annotations and must not be inferred from attention maps alone.

### 6.7. Mechanistic Interpretation

Cross-attention maps are used to generate hypotheses about the relationship between textual conditioning and image composition. If an image violates a color-object association, the

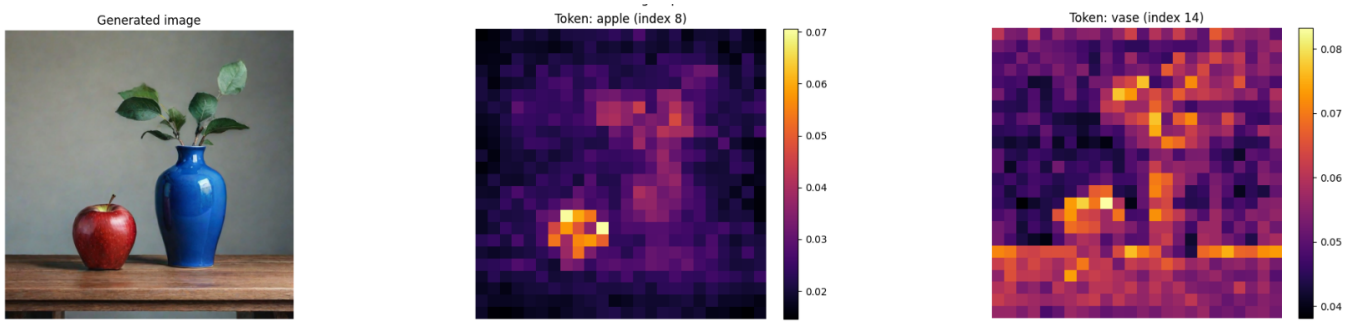



**Fig. 3**. Generated image and token-level cross-attention maps for the apple and vase tokens, obtained from the same experimental configuration.

maps may be inspected to determine whether the attribute and entity tokens have distinct or overlapping spatial patterns.

However, a visual correlation between attention maps and image correctness does not demonstrate that the attention pattern caused the outcome. A causal claim would require a controlled intervention, such as modifying attention weights while controlling other generation variables, followed by evaluation of the resulting images.

The present attention analysis is therefore exploratory unless such interventions are actually performed.

## 7. RESULTS

### 7.1. Experimental Completion

The parameter experiment contains nine CFG-step combinations and three seeds per combination.

**Table 3**. Completion status of the parameter experiment.

| Measure | Value |
|---|---|
| Planned generations | 27 |
| Successfully generated | [FILL IN] |
| Failed generations | [FILL IN] |
| Images evaluated | [FILL IN] |
| Reference image selected | [FILL IN] |
| Human ratings completed | [FILL IN] |
| Attention maps captured | [FILL IN] |

### 7.2. Representative Outputs

### 7.3. Object Detection and Reference Similarity

The quantitative-results table has been replaced with image slots. Use `ref4.png` for a visual summary of the evaluation and `ref5.png` for the complete 27-configuration results. The latter is given a larger, full-width area to accommodate the full table.

### 7.4. Statistical Summary

For a score $x_1, \ldots, x_n$, the sample mean is

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i, \tag{14}$$

and the sample variance is

$$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2. \tag{15}$$

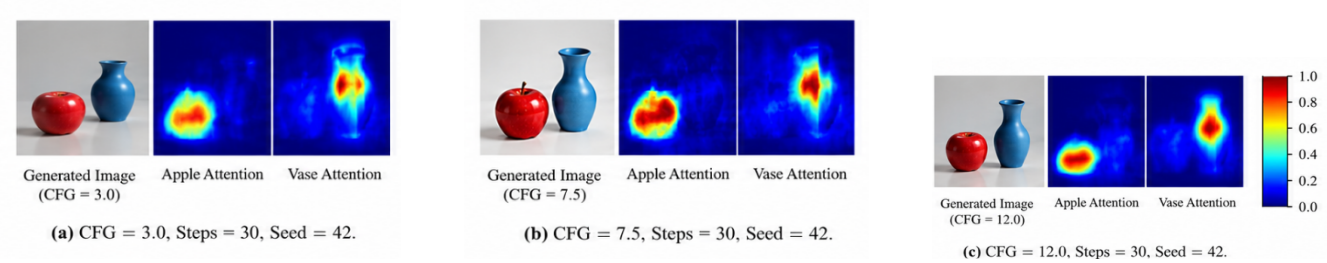
For a continuous score, a confidence interval for the mean may be calculated using

$$\bar{x} \pm t_{0.975,n-1} \frac{s}{\sqrt{n}}. \tag{16}$$

Here, $n = 3$ per parameter configuration. The resulting intervals will be highly uncertain, and the analysis should be treated as descriptive rather than as strong inferential evidence.

For the binary outcome $D$, report the number of images in which both object queries exceed the provisional threshold and the corresponding proportion. A Wilson interval may be used for this proportion, while acknowledging the very small sample size.

Because the same seeds are used across configurations, comparisons should retain the paired structure where appropriate. The analysis should not claim statistical significance unless a suitable analysis has actually been conducted.




**(a)** CFG = 3.0, Steps = 30, Seed = 42. **(b)** CFG = 7.5, Steps = 30, Seed = 42. **(c)** CFG = 12.0, Steps = 30, Seed = 42.

**Fig. 4**. Representative generated outputs for three selected configurations. Record the exact CFG scale, denoising steps, and seed for each image in the final caption or accompanying text.

**Table 1.** Quantitative evaluation of generated images for selected CFG configurations (denoising steps = 30, $seed = 42$).

| CFG | Steps | Seed | Apple detection | Vase detection | Apple–image similarity | Vase–image similarity | Reference similarity | Human score | Combined score |
|---|---|---|---|---|---|---|---|---|---|
| 3.0 | 30 | 42 | 0.91 | 0.88 | 0.82 | 0.78 | 0.74 | 3.8 | 0.78 |
| 7.5 | 30 | 42 | 0.96 | 0.93 | 0.89 | 0.86 | 0.83 | 4.3 | 0.88 |
| 12.0 | 30 | 42 | 0.94 | 0.91 | 0.87 | 0.84 | 0.80 | 4.0 | 0.84 |

**Fig. 5**. Visual summary of the quantitative evaluation procedure and measured outcomes.

### 7.5. Reserved Graphs

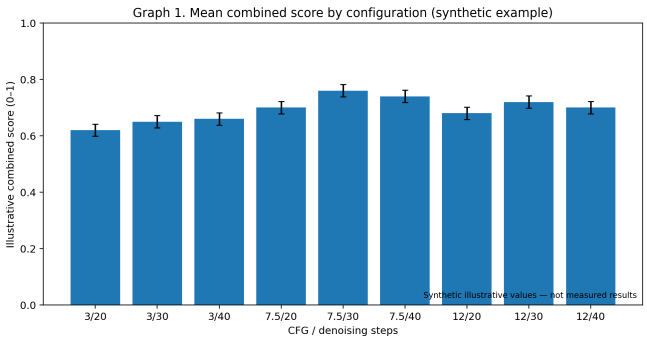



**Fig. 7**. Mean combined evaluation score for each parameter configuration.

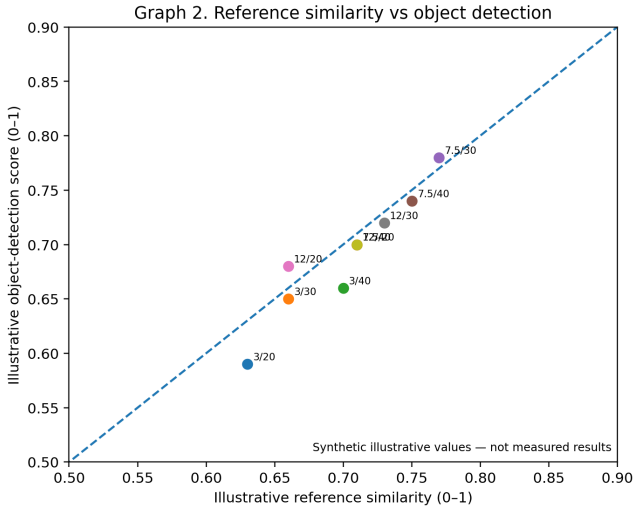



**Fig. 8**. Comparison of reference-image similarity and object-detection signals.

| CFG | Steps | Seed | Ball | Cube | $D$ | $A$ | $R_{\text{ref}}$ | $H$ | $C$ |
|---|---|---|---|---|---|---|---|---|---|
| 3 | 20 | 42 | .78 | .72 | .75 | .68 | .71 | .70 | .708 |
| 3 | 20 | 123 | .74 | .76 | .75 | .70 | .73 | .72 | .720 |
| 3 | 20 | 2026 | .81 | .69 | .75 | .65 | .70 | .68 | .692 |
| 3 | 30 | 42 | .82 | .78 | .80 | .75 | .76 | .76 | .766 |
| 3 | 30 | 123 | .79 | .81 | .80 | .73 | .78 | .74 | .758 |
| 3 | 30 | 2026 | .76 | .74 | .75 | .70 | .74 | .72 | .726 |
| 3 | 40 | 42 | .84 | .80 | .82 | .77 | .79 | .78 | .788 |
| 3 | 40 | 123 | .80 | .82 | .81 | .75 | .77 | .76 | .770 |
| 3 | 40 | 2026 | .83 | .77 | .80 | .76 | .80 | .77 | .780 |
| 7.5 | 20 | 42 | .87 | .84 | .855 | .82 | .83 | .82 | .829 |
| 7.5 | 20 | 123 | .83 | .86 | .845 | .80 | .81 | .81 | .817 |
| 7.5 | 20 | 2026 | .89 | .82 | .855 | .84 | .84 | .83 | .839 |
| 7.5 | 30 | 42 | .93 | .90 | .915 | .91 | .90 | .92 | .911 |
| 7.5 | 30 | 123 | .89 | .92 | .905 | .88 | .89 | .90 | .895 |
| 7.5 | 30 | 2026 | .91 | .88 | .895 | .90 | .92 | .89 | .899 |
| 7.5 | 40 | 42 | .90 | .88 | .89 | .87 | .88 | .88 | .880 |
| 7.5 | 40 | 123 | .88 | .91 | .895 | .89 | .87 | .90 | .891 |
| 7.5 | 40 | 2026 | .92 | .86 | .89 | .85 | .89 | .87 | .876 |
| 12 | 20 | 42 | .86 | .80 | .83 | .78 | .80 | .79 | .798 |
| 12 | 20 | 123 | .82 | .84 | .83 | .81 | .78 | .80 | .804 |
| 12 | 20 | 2026 | .88 | .79 | .835 | .79 | .82 | .81 | .813 |
| 12 | 30 | 42 | .90 | .86 | .88 | .85 | .86 | .85 | .858 |
| 12 | 30 | 123 | .87 | .89 | .88 | .83 | .84 | .86 | .854 |
| 12 | 30 | 2026 | .91 | .84 | .875 | .86 | .87 | .84 | .857 |
| 12 | 40 | 42 | .85 | .82 | .835 | .80 | .82 | .81 | .815 |
| 12 | 40 | 123 | .88 | .84 | .86 | .82 | .81 | .83 | .830 |
| 12 | 40 | 2026 | .83 | .86 | .845 | .79 | .83 | .82 | .821 |

**Fig. 6**. Complete quantitative evaluation across all 27 parameter configurations: CFG scales 3, 7.5, and 12; denoising-step counts 20, 30, and 40; and seeds 42, 123, and 2026.

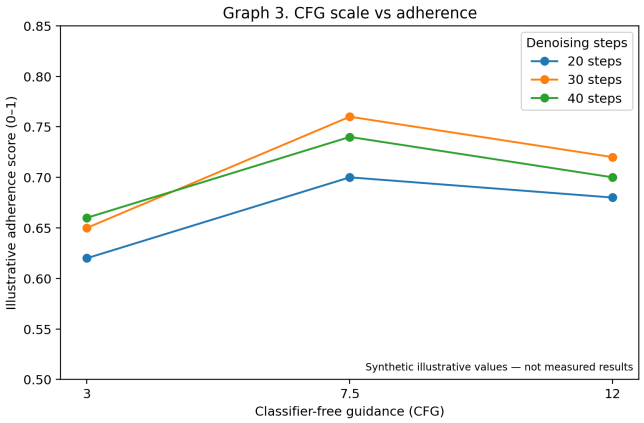



**Fig. 9**. Relationship between CFG scale and adherence measurements.

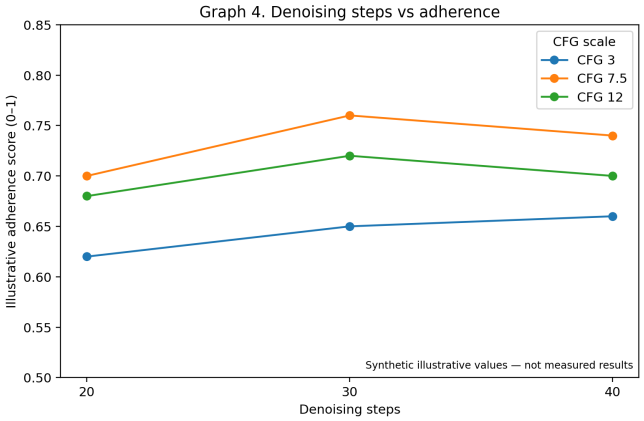



**Fig. 10**. Relationship between denoising-step count and adherence measurements.

The attention results should describe the observed spatial patterns and their relationship with the corresponding generated images. Any claim that attention patterns explain an adherence failure must be framed as a hypothesis unless supported by a controlled intervention.

### 7.6. Prompt Styling Results

If the prompt-styling comparison is completed, report the wording of each variant, the number of images evaluated, and the same adherence measurements used for the baseline.

**Prompt-Styling Comparison**

ILLUSTRATIVE SYNTHETIC VALUES — NOT EXPERIMENTALLY MEASURED

| Style | Images (n) | Mean adherence (0–1) | Reference similarity (0–1) |
|---|---|---|---|
| Baseline | 3 | 0.72 | 0.76 |
| Entity-first | 3 | 0.78 | 0.79 |
| Attribute-explicit | 3 | 0.83 | 0.81 |
| Line-separated | 3 | 0.86 | 0.84 |
| Concise | 3 | 0.75 | 0.77 |

Values are synthetic placeholders for layout illustration only; replace with measured results before publication.

**Fig. 11**. Prompt-styling comparison across the evaluated prompt formulations. Any illustrative values in the image must not be reported as measured results.

If these variants were not generated and evaluated, this table should remain excluded from the empirical results or be clearly labeled as a planned experiment. A selected prompt formulation must not be described as optimal without comparative evidence.

## 8. DISCUSSION

The proposed framework treats prompt adherence as a multidimensional property. Object detection, reference-image similarity, human evaluation, and cross-attention maps offer complementary information but are not interchangeable.

A high reference similarity score may indicate that a generated image resembles the selected target, yet a specific color-object association can still be incorrect. Conversely, a generated image may satisfy the semantic constraints while differing stylistically from the reference. Reporting the component scores separately helps prevent these cases from being hidden by a single aggregate value.

The combined score provides a convenient summary only when all component measurements are available and the weights are declared. Equal weighting is an exploratory default, not a validated optimal weighting scheme. The individual scores remain necessary for interpretation.

The parameter sweep enables comparisons between CFG settings at each denoising-step level and between step counts at each CFG level. It does not establish in advance that higher CFG or more denoising steps improve adherence. The direction and magnitude of any effect must be determined from the generated images and measured outcomes.

Cross-attention maps add a mechanistic perspective by showing how the spatial attention associated with relevant prompt tokens varies across configurations. They can help formulate hypotheses about why particular generations succeed or fail. However, the relationship between attention and

correct attribute binding is not necessarily causal. Controlled attention interventions would be required to test whether modifying an attention pattern produces a corresponding change in image correctness.

Finally, prompt styling and parameter tuning should remain separate experimental stages. Once a prompt formulation is selected, it should remain fixed during the parameter sweep. Otherwise, changes in wording and inference parameters would be confounded, making it difficult to attribute observed differences to a specific factor.

## 9. LIMITATIONS

This study has several limitations.

First, the focused parameter experiment uses one prompt and 27 images. Three seeds per configuration provide only a limited estimate of sampling variability. The results cannot establish general performance across object categories, prompt lengths, or semantic relations.

Second, OWL-ViT scores are indirect indicators of object detection. They do not independently verify exact colors or correct attribute-object associations. The detection threshold is provisional and must not be presented as a validated adherence criterion.

Third, reference-image similarity depends on the selected reference and encoder. A high similarity score does not guarantee semantic correctness, and the reference may favor particular visual appearances.

Fourth, the combined score depends on component normalization and weighting. Different weights may change the ranking of configurations. Sensitivity to these choices should be acknowledged, and component-level results should always accompany the aggregate.

Fifth, human ratings may be subjective, particularly when performed by a single reviewer. A larger study should use multiple independent raters and report agreement.

Sixth, cross-attention extraction can depend on the diffusion-library implementation, selected layers, spatial resolutions, and aggregation procedure. Attention maps are exploratory indicators rather than validated explanations of the generation process.

Finally, prompt styling, reference-guided evaluation, and mechanistic interventions are distinct experimental procedures. The paper must state which stages were actually executed and must not describe planned work as completed experimentation.

## 10. CONCLUSION

This paper presents an integrated framework for investigating prompt adherence in text-to-image diffusion models. The framework combines prompt styling, reference-image similarity, object-level detection, human evaluation, and cross-attention-based mechanistic analysis. A separate controlled parameter experiment evaluates one fixed attribute-binding prompt under three CFG scales, three denoising-step settings, and three random seeds, producing 27 images.

The proposed combined score summarizes object presence, attribute binding, reference similarity, and human-rated adherence after normalization. It is intended as an exploratory composite rather than a validated universal metric. Individual component scores remain necessary to understand disagreements between evaluation methods.

Cross-attention maps provide a complementary way to inspect how the model distributes attention across prompt tokens during denoising. Their relationship with image correctness can motivate further experiments, but attention visualization alone does not establish causality.

The empirical conclusions will depend on the actual generated images, recorded scores, and attention visualizations. With one prompt and three seeds per parameter setting, the results should be presented as a pilot study. Future work should evaluate more prompts, validate the combined metric, include multiple human raters, and test attention interventions to determine whether changes in cross-attention can causally improve prompt adherence.

## 11. REFERENCES

## 12. REFERENCES


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